

type of email, so that when recipients respond to your mail by hitting the 'Reply to' button on their email clients, they land in the boxes designated for them.

Publish the SPF and PTR records

An SPF or Sender Policy Framework specifies which servers can send emails to 'yourdomain.com'. In most cases, an SPF record is created by your hosting providers but for the sake of this article, let's create one for our bulk email server.

```
yourdomain.com. IN TXT "v=spf1 a mx ptr -all"
```

The `v=spf1` is the version specification for the SPF being used. 'a' means that the IP address of *yourdomain.com* is allowed to send emails while 'mx' means that even the 'mx' servers can send email. You can also specify a different server for 'mx' by mentioning 'mx:sastratechnologies.com'. The PTR mechanism means that if a host has a PTR record ending with *mydomain.com*, then it can send email.

Sign with DKIM using a key that's less than 1624 bits

DKIM stands for 'Domain Key Identified Mail'. It provides a method for validating a domain name identity that is associated with the message through cryptographic authentication. We'll use 'Opendkim Milter' to sign our outgoing email.

PHPList is an email generator and Opendkim is a milter or a mailfilter! The two must be bridged by an MTA or Mail Transfer Agent. Most distributions of Linux have Sendmail installed. We'll cover the process of installing DKIM and configuring Sendmail to use the DKIM milter. The process is similar to using Postfix as your MTA.

Install Opendkim by using the following command:

```
$yum install opendkim openssl
```

Set it to run on startup:

```
$chkconfig --levels 235 opendkim on
```

Generate a private key, as follows:

```
$openssl genrsa -out default.private 1024
```

Generate a public key:

```
openssl rsa -in default.private -pubout -out default.public -outform PEM
```

As the root user, or by using `sudo su`, move the private key to the directory for Opendkim to read the private key.

Ensure that you change the ownership of the key:

```
$mv default.private /etc/opendkim/keys/.
$chown opendkim /etc/opendkim/keys/default.private
```

We need to instruct Opendkim to use the KeyTable file. So, open the DKIM configuration file and look for the following line and uncomment it:

```
KeyTable /etc/opendkim/KeyTable
```

Create an entry in the KeyTable by editing the file using Nano or any other editor. The domain key should be as shown below:

```
$nano /etc/opendkim/KeyTable
```

```
default._domainkey.sastratechnologies.in sastratechnologies.in:default:/etc/opendkim/keys/default.private
```

Create a `TXT` record in your DNS Zone file, where you set the `HOST` to `default._domainkey` and the value to the key value. It should look something like what follows:

```
v=DKIM1; g=*; k=rsa; p=MIGfMA0GCQSqSib3DQEBAQUAA4GNADCBiQKBgQDrwLEIYP2BAC53zzJn3WpNc9oD
Mq8l+b+zs502FtjXVqi0+Fs4gHLJDCdlT0HpLU2hoPXpnyTFmi2/
lwsF9KXSGMLT
dy/XpiwaXYb1+Zw68PMwUougnF/XOkY8j2yTHF/9M+Zi0Rvnr6eGWLWTB
yJ07Y
Y7ED+Y0FYRCYqLq57WIDAQAB
```

...where 'p' is the value of the public key that we generated, i.e., `default.public`

Now instruct Sendmail to use DKIM by editing `submit.mc` and adding the following line:

```
nano /etc/mail/submit.mc
INPUT_MAIL_FILTER('dk-filter', 'S=inet:8891@localhost')dn1
```

Build a new `submit.mc` by running:

```
m4 /etc/mail/submit.mc > submit.cf
```

Restart Sendmail:

```
/etc/init.d/sendmail restart
```

Now all mails that you send out using PHPList will be signed by Sendmail using the OpenSSL keys we generated.

Update contact information on WHOIS and abuse.net

It's a good practice to have a listing on *abuse.net* to allow